Proof by Induction-Divisibility

Key Points

Remember to prove a statement holds for all positive integers by induction, we do the following:

- Show it holds for the base case 1
- Assume it holds for a case k
- Prove the statement then holds for the case $k+1$

A constant c can be shown to divide an expression $f(n)$, for any positive integer n using a proof by induction. For the inductive step, assume that $c \mid f(k)$ (c divides $f(k)$), and use this to show that $c \mid f(k+1)$

Basic Skills

Q1. What does the expression $3 \mid 9$ mean

Q2. Let $f(n) = 5n - 4$. Does 3 divide $f(n)$ for

a) $n = 1$ **b)** $n = 2$ **c)** $n = 3$ **d)** $n = 5$

Q3. Let $f(n) = 3n + 9$. Does $3 \mid f(n)$ for the following values of n

a) $n = 1$ **b)** $n = 2$ **c)** $n = 3$ **d)** $n = 5$

Q4. Suppose $3 \mid f(n)$ for all positive integers n. Consider the function $g(n) = f(n) + 9$. Show that $3 \mid g(n)$.

Q5. Suppose $2 \mid f(n)$ for all positive integers n. Consider the function $g(n) = f(n) + 2n^2 - 8$. Show that $2 \mid g(n)$.

Competency Skills

Q6. Consider the expression $f(n) = 11^n - 1$. We want to show that $5 \mid f(n)$ for all positive integers n using a proof by induction

a) Verify that the statement holds in the case $\mathrm{n} = 1$

b) Assume that $5 \mid f(k)$. Write this out as an equation with 11^k as the only term on the right hand side

c) Using b) show that $5 \mid f(k+1)$

d) Write out the statement we can conclude from parts a) and c) using proof by induction

Q7. Consider the expression $f(n) = n^3 + 5n$. Using a proof by induction show that $3 \mid f(n)$ for all positive integers n

Q8. Consider the expression $f(n) = 4^n + 6n - 1$. Using a proof by induction show that $9 \mid f(n)$ for all positive integers n

Advanced Skills

Q9. Show that the expression $(n+1)(n+3)$ is divisible by 4 for all odd positive integers.

Q10. Each month, a car salesman receives commission worth 4^{t+1} pounds, where t is the number of cars he has sold that month. As a Christmas bonus, he is given an additional 5^{2t-1} pounds for the month of December. Show that as long as he sells at least one car, he will always be able to split his December earnings evenly between himself, his wife, his parents and his three children.

Q11. Consider the expression $f(n) = 8^n + 1$

a) Show that if $7 \mid f(k)$ for some k, then $7 \mid f(k+1)$

b) Your friend concludes that $7 \mid f(n)$ for all positive integers n by induction. Why are they wrong?

Extension

Q12. Let k be any positive integer and consider the expression $f(n) = (2k+1)^n - (k+1)^n$. Using a proof by induction show that $k \mid f(n)$ for all positive integers n.